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Theoretical Physics

Interactions and Correlations in Condensed Matter

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Academic Year 2025/2026

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Introduction

1 | Second Quantization

In the first quantization formalism, we describe a system of N particles by a wavefunction $\Psi(\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_N)$, which is a function of the coordinates of all the particles. This wavefunction must satisfy the appropriate symmetry properties, depending on whether we are dealing with bosons or fermions. For bosons, the wavefunction must be symmetric under the exchange of any two particles, while for fermions, the wavefunction must be antisymmetric under the exchange of any two particles:

$$\Psi(\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_i, \dots, \mathbf{r}_j, \dots, \mathbf{r}_N) = \begin{cases} +\Psi(\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_j, \dots, \mathbf{r}_i, \dots, \mathbf{r}_N) & \text{for bosons,} \\ -\Psi(\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_j, \dots, \mathbf{r}_i, \dots, \mathbf{r}_N) & \text{for fermions.} \end{cases}$$

But an important limitation of this formalism is that it cannot describe systems with a variable number of particles, since the wavefunction is defined for a fixed number of particles

$$\Psi(\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_N) = \sum_{\nu_1 \dots \nu_N} B_{\nu_1 \dots \nu_N} S_{\pm} \phi_{\nu_1}(\mathbf{r}_1) \phi_{\nu_2}(\mathbf{r}_2) \cdots \phi_{\nu_N}(\mathbf{r}_N),$$

where $S_{\pm}$ is the symmetrization or antisymmetrization operator, depending on whether we are dealing with bosons or fermions, and $\phi_{\nu}(\mathbf{r})$ are the single-particle wavefunctions. The coefficients $B_{\nu_1 \dots \nu_N}$ are the expansion coefficients of the wavefunction in terms of the single-particle states. Thus, the wavefunction is defined for a fixed number of particles N , and it cannot describe processes that involve the creation or annihilation of particles. Furthermore we can write

$$S_{\pm} \phi_{\nu_1}(\mathbf{r}_1) \phi_{\nu_2}(\mathbf{r}_2) \cdots \phi_{\nu_N}(\mathbf{r}_N) = \begin{pmatrix} \psi_{\nu_1}(\mathbf{r}_1) & \cdots & \psi_{\nu_1}(\mathbf{r}_N) \\ \vdots & \ddots & \vdots \\ \psi_{\nu_N}(\mathbf{r}_1) & \cdots & \psi_{\nu_N}(\mathbf{r}_N) \end{pmatrix}_{\pm},$$

since the symmetrization or antisymmetrization operator can be expressed as a sum over all permutations of the particles, with a plus sign for bosons and a minus sign for fermions, which is equivalent to the determinant or permanent of the matrix of single-particle wavefunctions, depending on whether we are dealing with fermions or bosons.¹ Thus, the wavefunction is defined for a fixed number of particles N , and it cannot describe processes that involve the creation or annihilation of particles.

This is a problem when we want to describe systems that can exchange particles with their environment, such as in the case of open systems or in the case of systems that can undergo particle creation and annihilation, such as in quantum field theory. To overcome this limitation, we need to introduce a new formalism, which is called **second quantization**, which allows us to describe systems with a variable number of particles by introducing creation and annihilation operators.

¹The permanent of a matrix is the sum over all permutations of the matrix elements, with a plus sign for each permutation (while the determinant has alternating signs).

1.1 | Occupation Number Representation

In the context of second quantization, we can describe the state of a system of particles by specifying the **occupation numbers of the single-particle states** (seen as a set of quantum numbers). The occupation number n_ν of a single-particle state ν is defined as the number of particles that occupy that state.

Thus, we can represent the state of a N -particle system by a vector of occupation numbers:

$$|n_{\nu_1}, n_{\nu_2}, \dots, n_{\nu_m}\rangle, \quad \sum_{\nu} n_{\nu} = N,$$

where n_{ν_i} is the occupation number of the single-particle state ν_i , and the sum of all occupation numbers must be equal to the total number of particles N . This representation is called the **occupation number representation**, and it is a convenient way to describe many-body systems, since it allows us to keep track of the number of particles in each state without having to specify the coordinates of each particle. From this it follows straightforwardly the definition of the number operator $\hat{n}_\nu$, which counts the number of particles in the single-particle state ν :

$$\hat{n}_\nu |n_{\nu_1}, n_{\nu_2}, \dots, n_{\nu_m}\rangle = n_\nu |n_{\nu_1}, n_{\nu_2}, \dots, n_{\nu_m}\rangle.$$

For bosons, the occupation number can take any non-negative integer value, while for fermions, the occupation number can only be 0 or 1, due to the Pauli exclusion principle:

$$n_\nu = \begin{cases} 0, 1, 2, \dots & \text{for bosons,} \\ 0, 1 & \text{for fermions.} \end{cases}$$

In order to use this formalism, we need to introduce the analogue of the Hilbert space for the variable number of particles, which is called the **Fock space**. The Fock space is the direct sum of the Hilbert spaces for all possible numbers of particles:

$$\mathcal{F} = \bigoplus_{N=0}^{\infty} \mathcal{H}_N,$$

where $\mathcal{H}_N$ is the Hilbert space for N particles. The Fock space allows us to describe states with a variable number of particles, and it is the natural setting for the second quantization formalism. In the Fock space, we can define the vacuum state $|0\rangle$, which is the state with no particles, and we can build all other states by applying creation operators to the vacuum state. But before let us make an example.

Example (Fock space basis). Let us consider a system of bosons with varying number of particles, and let us choose a single-particle basis with two states, which we can label as $|1\rangle$ and $|2\rangle$. The Fock space for this system is the direct sum of the Hilbert spaces for all possible numbers of particles:

$$\mathcal{F} = \bigoplus_{N=0}^{\infty} \mathcal{H}_N.$$

We can show some examples of basis for the Fock space:

$$\begin{aligned} \mathcal{H}_0 &= |0, 0, \dots, 0\rangle, \\ \mathcal{H}_1 &= |1, 0, \dots, 0\rangle, |0, 1, \dots, 0\rangle, \dots, |0, 0, \dots, 1\rangle, \\ \mathcal{H}_2 &= |1, 1, 0, \dots, 0\rangle, \dots, |1, 0, \dots, 0, 1, 0, \dots, 0\rangle, |2, 0, \dots, 0\rangle, \dots \end{aligned}$$

where the first state is the vacuum state, the second states are the single-particle states, and the third states are the two-particle states. We can see that in the occupation number representation, we can easily keep track of the number of particles in each state, and we can build all possible states by permuting the occupation numbers.

Note that for a system of fermions, the basis states would be different, since the occupation numbers can only be 0 or 1, so we would not have states with more than one particle in the same state: the last state in the two-particle Hilbert space would not be allowed, since it would violate the Pauli exclusion principle.

Now it is time for us to introduce the creation and annihilation operators, which are the fundamental building blocks of the second quantization formalism.

1.1.1 | Ladder Operators

Ladder operators are operators that can raise or lower the occupation number of a single-particle state. We will separate the definition of the ladder operators for bosons and fermions, since they satisfy different commutation or anticommutation relations.

Bosons

For bosons, we define the **creation operator** $\hat{b}_\nu^\dagger$ and the **annihilation operator** $\hat{b}_\nu$ for a single-particle state ν as follows:

$$\begin{cases} \hat{b}_\nu |n_{\nu_1}, n_{\nu_2}, \dots, n_\nu, \dots\rangle = B_-(n_\nu) |n_{\nu_1}, n_{\nu_2}, \dots, n_\nu - 1, \dots\rangle, \\ \hat{b}_\nu^\dagger |n_{\nu_1}, n_{\nu_2}, \dots, n_\nu, \dots\rangle = B_+(n_\nu) |n_{\nu_1}, n_{\nu_2}, \dots, n_\nu + 1, \dots\rangle, \end{cases}$$

where $B_-(n_\nu)$ and $B_+(n_\nu)$ are coefficients that depend on the occupation number n_ν . The creation operator $\hat{b}_\nu^\dagger$ increases the occupation number of the state ν by one, while the annihilation operator $\hat{b}_\nu$ decreases the occupation number of the state ν by one. For bosons, these operators satisfy the following commutation relations:

$$\begin{cases} [\hat{b}_\nu, \hat{b}_{\nu'}^\dagger] = \delta_{\nu\nu'}, \\ [\hat{b}_\nu, \hat{b}_{\nu'}] = [\hat{b}_\nu^\dagger, \hat{b}_{\nu'}^\dagger] = 0, \end{cases}$$

where $[A, B] = AB - BA$ is the commutator of the operators A and B . The commutation relations reflect the fact that bosons can occupy the same state, and that the order of the operators does not matter.

Let us apply these operators to the occupation number states to find the coefficients $B_-(n_\nu)$ and $B_+(n_\nu)$, starting from the vacuum state $|0, 0, \dots, 0, \dots\rangle$:

$$\begin{aligned} \hat{b}_\nu |0, 0, \dots, 0, \dots\rangle &= 0, & B_-(0) &= 0, \\ \hat{b}_\nu^\dagger |0, 0, \dots, 0, \dots\rangle &= |0, 0, \dots, 0, 1, 0, \dots\rangle, & B_+(0) &= 1, \end{aligned}$$

where we understood how the annihilation operator gives zero when applied to the vacuum state, while the creation operator creates a particle in the state ν .

Now let us apply the operators to a generic normalized occupation number state $|\phi\rangle$:

$$\hat{b}_\nu |\phi\rangle, \quad \langle\phi| \hat{b}_\nu^\dagger \hat{b}_\nu |\phi\rangle = |\hat{b}_\nu |\phi\rangle| = \lambda > 0,$$

such that if we apply the creation operator to the state $\hat{b}_\nu |\phi\rangle$ we get:

$$\hat{b}_\nu^\dagger \hat{b}_\nu |\phi\rangle = \lambda |\phi\rangle = n_\nu |\phi\rangle,$$

since the number operator $\hat{n}_\nu = \hat{b}_\nu^\dagger \hat{b}_\nu$ counts the number of particles in the state ν . Now if we apply the number operator $\hat{n}_\nu = \hat{b}_\nu^\dagger \hat{b}_\nu$ to the state $\hat{b}_\nu |\phi\rangle$, using the commutation relations, we should be able to find the coefficient $B_-(n_\nu)$:

$$\hat{n}_\nu \hat{b}_\nu |\phi\rangle = (\hat{b}_\nu^\dagger \hat{b}_\nu) \hat{b}_\nu |\phi\rangle = (\hat{b}_\nu \hat{b}_\nu^\dagger - 1) \hat{b}_\nu |\phi\rangle = \hat{b}_\nu (\hat{b}_\nu^\dagger \hat{b}_\nu - 1) |\phi\rangle = (\lambda - 1) \hat{b}_\nu |\phi\rangle,$$

so that we can identify the coefficient $B_-(n_\nu)$ as:

$$B_-(n_\nu) = \sqrt{n_\nu},$$

and by applying the number operator to the state $\hat{b}_\nu^\dagger |\phi\rangle$, we get:

$$\hat{n}_\nu \hat{b}_\nu^\dagger |\phi\rangle = (\hat{b}_\nu^\dagger \hat{b}_\nu) \hat{b}_\nu^\dagger |\phi\rangle = \hat{b}_\nu^\dagger (\hat{b}_\nu \hat{b}_\nu^\dagger) |\phi\rangle = \hat{b}_\nu^\dagger (\hat{b}_\nu^\dagger \hat{b}_\nu + 1) |\phi\rangle = (\lambda + 1) \hat{b}_\nu^\dagger |\phi\rangle,$$

so that we can identify the coefficient $B_+(n_\nu)$ as:

$$B_+(n_\nu) = \sqrt{n_\nu + 1}.$$

Thus, the action of the creation and annihilation operators on the occupation number states can be expressed as:

$$\begin{cases} \hat{b}_\nu |n_{\nu_1}, n_{\nu_2}, \dots, n_\nu, \dots\rangle = \sqrt{n_\nu} |n_{\nu_1}, n_{\nu_2}, \dots, n_\nu - 1, \dots\rangle, \\ \hat{b}_\nu^\dagger |n_{\nu_1}, n_{\nu_2}, \dots, n_\nu, \dots\rangle = \sqrt{n_\nu + 1} |n_{\nu_1}, n_{\nu_2}, \dots, n_\nu + 1, \dots\rangle. \end{cases}$$

Fermions

For fermions, we define the **creation operator** $\hat{c}_\nu^\dagger$ and the **annihilation operator** $\hat{c}_\nu$ for a single-particle state ν with the same action on the occupation number states as for bosons, but with the coefficients $B_-(n_\nu)$ and $B_+(n_\nu)$ that depend on the occupation number n_ν in a different way, since for fermions the occupation number can only be 0 or 1:

$$\begin{cases} \hat{c}_\nu |n_{\nu_1}, n_{\nu_2}, \dots, n_\nu, \dots\rangle = C_-(n_\nu) |n_{\nu_1}, n_{\nu_2}, \dots, n_\nu - 1, \dots\rangle, \\ \hat{c}_\nu^\dagger |n_{\nu_1}, n_{\nu_2}, \dots, n_\nu, \dots\rangle = C_+(n_\nu) |n_{\nu_1}, n_{\nu_2}, \dots, n_\nu + 1, \dots\rangle, \end{cases}$$

with

$$C_-(n_\nu) = \begin{cases} 0 & \text{if } n_\nu = 0, \\ 1 & \text{if } n_\nu = 1, \end{cases} \quad C_+(n_\nu) = \begin{cases} 1 & \text{if } n_\nu = 0, \\ 0 & \text{if } n_\nu = 1, \end{cases}$$

so that the creation operator $\hat{c}_\nu^\dagger$ creates a particle in the state ν if it is not already occupied, while the annihilation operator $\hat{c}_\nu$ annihilates a particle in the state ν if it is occupied. For fermions, these operators satisfy the following anticommutation relations:

$$\begin{cases} \{\hat{c}_\nu, \hat{c}_\nu^\dagger\} = 1, \\ \{\hat{c}_\nu, \hat{c}_\nu\} = \{\hat{c}_\nu^\dagger, \hat{c}_\nu^\dagger\} = 0, \end{cases}$$

since the multiparticle wavefunction must be antisymmetric under the exchange of any two particles, and thus the order of the operators matters. The anticommutation relations reflect the fact that fermions cannot occupy the same state:

$$\hat{c}_\nu^\dagger \hat{c}_\nu^\dagger = \frac{1}{2} \{\hat{c}_\nu^\dagger, \hat{c}_\nu^\dagger\} = 0,$$

and similarly for the annihilation operator.

1.2 | Hamiltonian of an Interacting System

In a many-body system, the Hamiltonian has to contain both a kinetic term, which describes the motion of the particles, and an interaction term, which describes the interactions between the particles. Let us show how to write the Hamiltonian of an interacting system in the second quantization formalism, starting from the first quantization formalism. We will see how many results from first quantization come in handy when it will be time to build the second quantization version.

First Quantization

In first quantization, the Hamiltonian of a system of N interacting particles can be written as:

$$H = \sum_{i=1}^N \hat{T}(\mathbf{r}_i, \nabla_{\mathbf{r}_i}) + \frac{1}{2} \sum_{i \neq j} \hat{V}(\mathbf{r}_i - \mathbf{r}_j),$$

where $\hat{T}(\mathbf{r}_i, \nabla_{\mathbf{r}_i})$ is the **one-body operator** for the i -th particle, and $\hat{V}(\mathbf{r}_i - \mathbf{r}_j)$ is the interaction potential, or **two body operator**, between the i -th and j -th particles. The factor of $\frac{1}{2}$ is included to avoid double counting the interactions between pairs of particles.

The easiest case in which we can write the Hamiltonian of a many-body system is the Helium atom.

Example (Helium atom). Let us consider the helium atom, which consists of two electrons interacting with each other and with the nucleus. The Hamiltonian of the helium atom can be written as:

$$H = \left[-\frac{\hbar^2}{2m} \nabla_{\mathbf{r}_1}^2 - \frac{Ze^2}{4\pi\epsilon_0 r_1} \right] + \left[-\frac{\hbar^2}{2m} \nabla_{\mathbf{r}_2}^2 - \frac{Ze^2}{4\pi\epsilon_0 r_2} \right] + \frac{e^2}{4\pi\epsilon_0 |\mathbf{r}_1 - \mathbf{r}_2|},$$

where m is the mass of the electron, Z is the atomic number of the nucleus (which is 2 for helium), e is the elementary charge, ϵ_0 is the vacuum permittivity and r_i is the distance of the i -th electron from the nucleus. The first term in square brackets represent the contribution of the kinetic energy and the potential energy of the first electron, while the second term in square brackets for the second electron; the last term represent the interaction between the two electrons, which is the **electron-electron interaction**. This Hamiltonian is a good starting point for understanding the electronic structure of the helium atom, and it can be solved using various approximation methods, such as perturbation theory or variational methods.

Let us study more in detail the structure of the one-body and two-body operators.

One-body operator. The one-body operator $\hat{T}(\mathbf{r}_j, \nabla_{\mathbf{r}_j})$ is an operator that acts on the coordinates of a single particle, and it typically includes the kinetic energy term and the potential energy term due to external fields. It can be expressed in terms of the single-particle states as:

$$T_{\nu_a \nu_b} = \int d^3\mathbf{r}_j \Psi_{\nu_a}^*(\mathbf{r}_j) \hat{T}(\mathbf{r}_j, \nabla_{\mathbf{r}_j}) \Psi_{\nu_b}(\mathbf{r}_j),$$

where $\Psi_{\nu}(\mathbf{r}_j)$ are the single-particle wavefunctions, and the integral is taken over all space. The internal operator $\hat{T}(\mathbf{r}_j, \nabla_{\mathbf{r}_j})$ can be expressed as the sum of the kinetic energy operator and the potential energy operator:

$$\hat{T}(\mathbf{r}_j, \nabla_{\mathbf{r}_j}) = -\frac{\hbar^2}{2m} \nabla_{\mathbf{r}_j}^2 + V_{\text{ext}}(\mathbf{r}_j),$$

and since $T_{\nu_a \nu_b}$ is the matrix element of the one-body operator between the single-particle states ν_a and ν_b , we can write:

$$\hat{T}(\mathbf{r}_j, \nabla_{\mathbf{r}_j}) = \sum_{\nu_a \nu_b} T_{\nu_a \nu_b} |\Psi_{\nu_b}(\mathbf{r}_j)\rangle \langle \Psi_{\nu_a}(\mathbf{r}_j)|,$$

where $|\Psi_{\nu_a}(\mathbf{r}_j)\rangle$ and $|\Psi_{\nu_b}(\mathbf{r}_j)\rangle$ are the single-particle states corresponding to the quantum numbers ν_a and ν_b , respectively.

Two-body operator. The two-body operator $\hat{V}(\mathbf{r}_j - \mathbf{r}_k)$ is an operator that acts on the coordinates of two particles, and it typically includes the interaction potential between the particles. It can be expressed in terms of the single-particle states as:

$$V_{\nu_c \nu_d, \nu_a \nu_b} = \int d^3 \mathbf{r}_j d^3 \mathbf{r}_k \Psi_{\nu_c}^*(\mathbf{r}_j) \Psi_{\nu_d}^*(\mathbf{r}_k) \hat{V}(\mathbf{r}_j - \mathbf{r}_k) \Psi_{\nu_a}(\mathbf{r}_j) \Psi_{\nu_b}(\mathbf{r}_k),$$

where again $\Psi_{\nu}(\mathbf{r}_j)$ are the single-particle wavefunctions, and the integral is taken over all space. The internal operator $\hat{V}(\mathbf{r}_j - \mathbf{r}_k)$ can then be expressed as a sum over the single-particle states as:

$$\hat{V}(\mathbf{r}_j - \mathbf{r}_k) = \sum_{\nu_a \nu_b \nu_c \nu_d} V_{\nu_c \nu_d, \nu_a \nu_b} |\Psi_{\nu_c}(\mathbf{r}_j)\rangle |\Psi_{\nu_d}(\mathbf{r}_k)\rangle \langle \Psi_{\nu_b}(\mathbf{r}_k)| \langle \Psi_{\nu_a}(\mathbf{r}_j)|,$$

where $V_{\nu_c \nu_d, \nu_a \nu_b}$ are the matrix elements of the two-body operator between the single-particle states.

Now that we have expressed the one-body and two-body operators in terms of the single-particle states, we can rewrite the Hamiltonian of the system in terms of these operators as:

$$H = \sum_{i=1}^N \sum_{\nu_a \nu_b} T_{\nu_a \nu_b} |\Psi_{\nu_b}(\mathbf{r}_i)\rangle \langle \Psi_{\nu_a}(\mathbf{r}_i)| + \frac{1}{2} \sum_{i \neq j} \sum_{\nu_a \nu_b \nu_c \nu_d} V_{\nu_c \nu_d, \nu_a \nu_b} |\Psi_{\nu_c}(\mathbf{r}_i)\rangle |\Psi_{\nu_d}(\mathbf{r}_j)\rangle \langle \Psi_{\nu_b}(\mathbf{r}_j)| \langle \Psi_{\nu_a}(\mathbf{r}_i)|,$$

where we have expressed the Hamiltonian in terms of the matrix elements of the one-body and two-body operators between the single-particle states. This Hamiltonian is still in a first quantization form, since it is expressed in terms of the coordinates of the particles, but it is a convenient starting point for expressing the Hamiltonian in terms of the creation and annihilation operators in the second quantization formalism.

Second Quantization

In order to express the Hamiltonian in terms of the creation and annihilation operators, we need to understand how to express the operators in second quantization. The key idea is to express the operators in terms of the ladder operators, which can be done by using the completeness relation of the single-particle states.

It can be shown that an operator in first quantization can be expressed in second quantization as:

$$\hat{A} = \sum_{\nu_a \nu_b} A_{\nu_a \nu_b} \hat{a}_{\nu_a}^\dagger \hat{a}_{\nu_b},$$

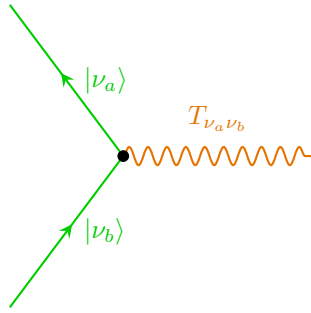
where $A_{\nu_a \nu_b}$ are the matrix elements of the operator in the single-particle basis, and $\hat{a}_{\nu}^\dagger$ and $\hat{a}_{\nu}$ are the creation and annihilation operators for the single-particle state ν , respectively. Thus there exists a systematic way to express any operator in first quantization in terms of the creation and annihilation operators in second quantization, by expressing the operator in terms of its

matrix elements in the single-particle basis, and then replacing the single-particle states with the corresponding creation and annihilation operators.

We can then write the Hamiltonian of the system in second quantization as:

$$H = \sum_{\nu_a \nu_b} T_{\nu_a \nu_b} \hat{a}_{\nu_a}^\dagger \hat{a}_{\nu_b} + \frac{1}{2} \sum_{\nu_a \nu_b \nu_c \nu_d} V_{\nu_c \nu_d, \nu_a \nu_b} \hat{a}_{\nu_c}^\dagger \hat{a}_{\nu_d}^\dagger \hat{a}_{\nu_a} \hat{a}_{\nu_b},$$

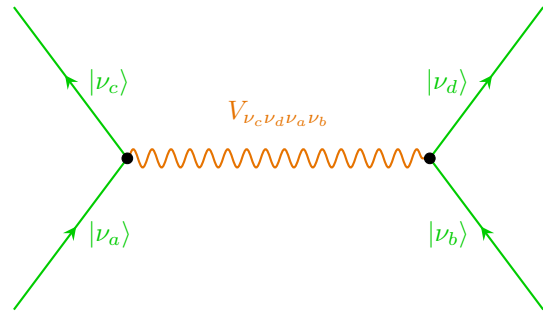
where the first term represents the one-body interactions, and the second term represents the two-body interactions between the particles. This Hamiltonian is now expressed in terms of the creation and annihilation operators, and it can be used to describe systems with a variable number of particles, since the creation and annihilation operators can create or annihilate particles in the system.



In the diagram on the left, we can see the one-body interaction between the single-particle states $|\nu_a\rangle$ and $|\nu_b\rangle$, which is represented by the matrix element $T_{\nu_a \nu_b}$. The particle represented by the line entering the vertex is in the state $|\nu_b\rangle$, while the particle represented by the line exiting the vertex is in the state $|\nu_a\rangle$. The interaction is represented by a wavy line, which indicates that it is a one-body interaction, since it involves only one particle.

The vertex represents the point where the interaction takes place, and it is associated with the matrix element $T_{\nu_a \nu_b}$, which quantifies the strength of the interaction between the two states, or else the probability amplitude for the transition from state $|\nu_b\rangle$ to state $|\nu_a\rangle$ due to the one-body operator. Basically we are annihilating a particle in the state $|\nu_b\rangle$ and creating a particle in the state $|\nu_a\rangle$ due to the one-body interaction.

For the two body interaction, represented in the diagram on the right, we have two vertices representing the interaction between the single-particle states $|\nu_a\rangle$, $|\nu_b\rangle$, $|\nu_c\rangle$ and $|\nu_d\rangle$. The particles represented by the lines entering the vertices are in the states $|\nu_a\rangle$ and $|\nu_b\rangle$, while the particles represented by the lines exiting the vertices are in the states $|\nu_c\rangle$ and $|\nu_d\rangle$.



The interaction is represented by a wavy line connecting the two vertices, which indicates that it is a two-body interaction, since it involves two particles. The matrix element $V_{\nu_c \nu_d, \nu_a \nu_b}$ quantifies the strength of the interaction between the four states, or else the probability amplitude for the transition from states $|\nu_a\rangle$ and $|\nu_b\rangle$ to states $|\nu_c\rangle$ and $|\nu_d\rangle$ due to the two-body operator. We are describing a process in which we are annihilating two particles in the states $|\nu_a\rangle$ and $|\nu_b\rangle$ and creating two particles in the states $|\nu_c\rangle$ and $|\nu_d\rangle$ due to the two-body interaction.

1.3 | Field Operators

Field operators are a powerful tool in quantum mechanics, particularly in the study of many-body systems. They allow us to describe the creation and annihilation of particles in a given state. The field operator $\hat{\Psi}(\mathbf{r})$ can be expressed as a sum over a complete set of single-particle states:

$$\begin{aligned}\hat{\Psi}(\mathbf{r}) &= \sum_{\nu} \phi_{\nu}(\mathbf{r}) \hat{a}_{\nu}, \\ \hat{\Psi}^{\dagger}(\mathbf{r}) &= \sum_{\nu} \phi_{\nu}^*(\mathbf{r}) \hat{a}_{\nu}^{\dagger},\end{aligned}$$

If we have an homogeneous system, we can choose plane waves as our single-particle states to sum only over momentum states:

$$\begin{aligned}\hat{\Psi}(\mathbf{r}) &= \frac{1}{\sqrt{\Omega}} \sum_{\mathbf{k}} e^{-i\mathbf{k}\cdot\mathbf{r}} \hat{a}_{\mathbf{k}}, \\ \hat{\Psi}^{\dagger}(\mathbf{r}) &= \frac{1}{\sqrt{\Omega}} \sum_{\mathbf{k}} e^{i\mathbf{k}\cdot\mathbf{r}} \hat{a}_{\mathbf{k}}^{\dagger},\end{aligned}$$

we are summing over all quantum numbers ν , which in this case are the momentum states $\mathbf{k}$, then we are creating or annihilating particles in all of those momentum states for a precise position $\mathbf{r}$. Since this operators are based on the same annihilation and creation operators, they satisfy the same commutation or anticommutation relations, depending on whether we are dealing with bosons or fermions. For bosons, the field operators satisfy the commutation relations:

$$\begin{cases} [\hat{\Psi}(\mathbf{r}), \hat{\Psi}^{\dagger}(\mathbf{r}')] = \delta(\mathbf{r} - \mathbf{r}'), \\ [\hat{\Psi}(\mathbf{r}), \hat{\Psi}(\mathbf{r}')] = [\hat{\Psi}^{\dagger}(\mathbf{r}), \hat{\Psi}^{\dagger}(\mathbf{r}')] = 0, \end{cases}$$

while for fermions, they satisfy the anticommutation relations:

$$\begin{cases} \{\hat{\Psi}(\mathbf{r}), \hat{\Psi}^{\dagger}(\mathbf{r}')\} = \delta(\mathbf{r} - \mathbf{r}'), \\ \{\hat{\Psi}(\mathbf{r}), \hat{\Psi}(\mathbf{r}')\} = \{\hat{\Psi}^{\dagger}(\mathbf{r}), \hat{\Psi}^{\dagger}(\mathbf{r}')\} = 0. \end{cases}$$

Instead of wavefunctions, we can also express the field operators in terms of the position basis.

Rewriting the Hamiltonian of our problem:

$$H = \sum_{\nu_a, \nu_b} \hat{T}_{\nu_a \nu_b} \hat{a}_{\nu_a}^{\dagger} \hat{a}_{\nu_b} + \frac{1}{2} \sum_{\nu_a, \nu_b, \nu_c, \nu_d} \hat{V}_{\nu_c \nu_d \nu_a \nu_b} \hat{a}_{\nu_c}^{\dagger} \hat{a}_{\nu_d}^{\dagger} \hat{a}_{\nu_a} \hat{a}_{\nu_b},$$

where we are summing over all quantum numbers ν , we can express the Hamiltonian in terms of the field operators as:

$$\begin{aligned}H &= \sum_{\nu_a, \nu_b} \left(\int d^3\mathbf{r} \Psi_{\nu_a}^*(\mathbf{r}) T(\mathbf{r}, \nabla_{\mathbf{r}}) \Psi_{\nu_b}(\mathbf{r}) \right) \hat{a}_{\nu_a}^{\dagger} \hat{a}_{\nu_b} + \frac{1}{2} \sum_{\nu_a, \nu_b, \nu_c, \nu_d} \left(\int d^3\mathbf{r} d^3\mathbf{r}' \Psi_{\nu_c}^*(\mathbf{r}) \Psi_{\nu_d}^*(\mathbf{r}') V(\mathbf{r} - \mathbf{r}') \Psi_{\nu_a}(\mathbf{r}) \Psi_{\nu_b}(\mathbf{r}') \right) \hat{a}_{\nu_c}^{\dagger} \hat{a}_{\nu_d}^{\dagger} \hat{a}_{\nu_a} \hat{a}_{\nu_b} \\ &\quad [\dots]\end{aligned}$$

Kinetic term. Let us analyze the kinetic term of the Hamiltonian:

- in **first quantization**, the kinetic term is given by:

$$\hat{T}(\mathbf{r}, \nabla_{\mathbf{r}}) = -\frac{\hbar^2}{2m} \nabla_{\mathbf{r}}^2 \delta_{\sigma\sigma'},$$

where we are using the basis $|\mathbf{k}, \sigma\rangle$, such that

$$\langle \mathbf{r} | \mathbf{k}, \sigma \rangle = \frac{1}{\sqrt{\Omega}} e^{i\mathbf{k} \cdot \mathbf{r}},$$

and if we compute the matrix element of the kinetic term, we get:

$$\langle \mathbf{k}_a, \sigma_a | \hat{T}(\mathbf{r}, \nabla_{\mathbf{r}}) | \mathbf{k}_b, \sigma_b \rangle = \frac{\hbar^2 k_b^2}{2m} \delta_{\mathbf{k}_a \mathbf{k}_b} \delta_{\sigma_a \sigma_b},$$

- in **second quantization**, the kinetic term can be expressed as:

$$T = \sum_{\nu_a, \nu_b} T_{\nu_a \nu_b} \hat{a}_{\nu_a}^\dagger \hat{a}_{\nu_b} = \sum_{\mathbf{k}, \mathbf{k}', \sigma, \sigma'} \langle \mathbf{k}, \sigma | \hat{T}(\mathbf{r}, \nabla_{\mathbf{r}}) | \mathbf{k}', \sigma' \rangle \hat{a}_{\mathbf{k}, \sigma}^\dagger \hat{a}_{\mathbf{k}', \sigma'} = \sum_{\mathbf{k}, \sigma} \frac{\hbar^2 k^2}{2m} \hat{a}_{\mathbf{k}, \sigma}^\dagger \hat{a}_{\mathbf{k}, \sigma},$$

Coulomb term. The Coulomb term of the Hamiltonian can be expressed as:

$$\begin{aligned} V &= \frac{1}{2} \sum_{\sigma, \sigma'} \sum_{\mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4} \langle \mathbf{k}_3, \sigma; \mathbf{k}_4, \sigma' | V(\mathbf{r} - \mathbf{r}') | \mathbf{k}_2, \sigma'; \mathbf{k}_1, \sigma \rangle \hat{a}_{\mathbf{k}_3, \sigma}^\dagger \hat{a}_{\mathbf{k}_4, \sigma'}^\dagger \hat{a}_{\mathbf{k}_2, \sigma'} \hat{a}_{\mathbf{k}_1, \sigma} \\ &= \frac{1}{2} \sum_{\sigma, \sigma'} \sum_{\mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4} \left[\int \int d^3\mathbf{r} d^3\mathbf{r}' \frac{e^{-i\mathbf{k}_3 \cdot \mathbf{r}}}{\sqrt{\Omega}} \frac{e^{-i\mathbf{k}_4 \cdot \mathbf{r}'}}{\sqrt{\Omega}} V(\mathbf{r} - \mathbf{r}') \frac{e^{i\mathbf{k}_2 \cdot \mathbf{r}'}}{\sqrt{\Omega}} \frac{e^{i\mathbf{k}_1 \cdot \mathbf{r}}}{\sqrt{\Omega}} \right] \hat{a}_{\mathbf{k}_3, \sigma}^\dagger \hat{a}_{\mathbf{k}_4, \sigma'}^\dagger \hat{a}_{\mathbf{k}_2, \sigma'} \hat{a}_{\mathbf{k}_1, \sigma} \\ &= \frac{1}{2} \sum_{\sigma, \sigma'} \sum_{\mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4} \frac{e^2}{\Omega^2} \left[\int \int d^3\mathbf{r} d^3\mathbf{r}' \frac{e^{i(\mathbf{k}_1 - \mathbf{k}_3) \cdot \mathbf{r}} e^{i(\mathbf{k}_2 - \mathbf{k}_4) \cdot \mathbf{r}'}}{|\mathbf{r} - \mathbf{r}'|} \right] \hat{a}_{\mathbf{k}_3, \sigma}^\dagger \hat{a}_{\mathbf{k}_4, \sigma'}^\dagger \hat{a}_{\mathbf{k}_2, \sigma'} \hat{a}_{\mathbf{k}_1, \sigma} \end{aligned}$$

if we now perform the change of variables $\boldsymbol{\xi} = \mathbf{r} - \mathbf{r}'$ we can express the Coulomb term as:

$$= \frac{1}{2} \sum_{\sigma, \sigma'} \sum_{\mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4} \frac{e^2}{\Omega^2} \left[\int d^3\mathbf{r} e^{i(\mathbf{k}_1 - \mathbf{k}_3) \cdot \mathbf{r}} \int d^3\boldsymbol{\xi} e^{i(\mathbf{k}_2 - \mathbf{k}_4) \cdot (\mathbf{r} - \boldsymbol{\xi})} \frac{1}{|\boldsymbol{\xi}|} \right] \hat{a}_{\mathbf{k}_3, \sigma}^\dagger \hat{a}_{\mathbf{k}_4, \sigma'}^\dagger \hat{a}_{\mathbf{k}_2, \sigma'} \hat{a}_{\mathbf{k}_1, \sigma}.$$

TODO: correct computations

So that in the end we find the Fourier transform of the Coulomb potential:

$$\tilde{V} = \int d^3\boldsymbol{\xi} \frac{e^{i(\mathbf{k}_2 - \mathbf{k}_4) \cdot \boldsymbol{\xi}}}{|\boldsymbol{\xi}|} = \frac{4\pi}{|\mathbf{k}_2 - \mathbf{k}_4|^2},$$

as we can see, the Coulomb potential is a very large range potential, since its transform as $\mathbf{q} \rightarrow 0$ diverges:

$$\tilde{V} = \frac{4\pi}{|\mathbf{q}|^2}.$$

In the end, integrating over the delta functions, we can express the Coulomb term as:

$$V = \frac{1}{2\Omega} \sum_{\sigma, \sigma'} \sum_{\mathbf{k}_1, \mathbf{k}_2, \mathbf{q}} \frac{4\pi e^2}{|\mathbf{q}|^2} \hat{a}_{\mathbf{k}_1 + \mathbf{q}, \sigma}^\dagger \hat{a}_{\mathbf{k}_2 - \mathbf{q}, \sigma'}^\dagger \hat{a}_{\mathbf{k}_2, \sigma'} \hat{a}_{\mathbf{k}_1, \sigma}.$$

Thus during the interaction, two particles with momenta $\mathbf{k}_1$ and $\mathbf{k}_2$ exchange a momentum $\mathbf{q}$ and end up with momenta $\mathbf{k}_1 + \mathbf{q}$ and $\mathbf{k}_2 - \mathbf{q}$, so that the total momentum is conserved during the interaction.² This is a consequence of the translational invariance of the system, which implies that the total momentum is a good quantum number. For the spin part, we know that the Coulomb interaction does not depend on the spin of the particles,³ so that the spin is also conserved during the interaction, which is a consequence of the rotational invariance of the system.

TODO: add drawing of the interaction (ie feynman diagram).

Thus we have modeled the interaction between the particles as a two-body interaction, which then scatters two particles with momenta $\mathbf{k}_1$ and $\mathbf{k}_2$ into two particles with momenta $\mathbf{k}_1 + \mathbf{q}$ and $\mathbf{k}_2 - \mathbf{q}$,

²The momentum conservation is enforced at the end by the delta function $\delta_{\mathbf{k}_1, \mathbf{k}_3 + \mathbf{q}}$.

³In some kind of interactions, we have other terms including all the Pauli matrices, so spin can switch.

where the momentum $\mathbf{q}$ is the momentum exchanged during the interaction. This is a very general form of the interaction, which can be applied to any system with translational invariance, and it is the starting point for many-body perturbation theory and other techniques used to study interacting systems.

This Hamiltonian cannot be solved exactly (for more than two particles), so we need to use some approximation methods to study the properties of the system. We are so going to develop the many body perturbation theory in the next chapters, which is a powerful tool to study interacting systems, and it is based on the idea of treating the interaction as a perturbation to the non-interacting system.

1.4 | The Electron Gas

The electron gas is a model of a system of electrons that are free to move in a uniform background of positive charge, which is often used to study the properties of metals and other materials. This model is also known as the **jellium model**, and it is a useful tool to understand the behavior of electrons in a solid. Its main features are:

- The electrons are treated as free particles, which means that they do not interact with each other, but they do interact with the positive background charge.
- The positive background charge is treated as a uniform distribution of charge, which means that it does not have any structure or dynamics.
- The system is assumed to be infinite and homogeneous, which means that it has no boundaries and that the properties of the system are the same everywhere.

The Hamiltonian of the electron gas can be expressed as:

$$H = (T_{\text{ion}} + V_{\text{ion-ion}}) + (T_{\text{el}} + V_{\text{el-el}}) + V_{\text{el-ion}},$$

where T_{ion} is the kinetic energy of the ions, $V_{\text{ion-ion}}$ is the potential energy of the ion-ion interaction, T_{el} is the kinetic energy of the electrons, $V_{\text{el-el}}$ is the potential energy of the electron-electron interaction, and $V_{\text{el-ion}}$ is the potential energy of the electron-ion interaction. Let us review term by term, starting from the ionic part:⁴

$$T_{\text{ion}} + V_{\text{ion-ion}} = \int d^3\mathbf{R} \hat{\Psi}_{\text{ion}}^\dagger(\mathbf{R}) \left(-\frac{\hbar^2}{2M} \nabla_{\mathbf{R}}^2 \right) \hat{\Psi}_{\text{ion}}(\mathbf{R}) + \frac{1}{2} \int \int d^3\mathbf{R}_1 d^3\mathbf{R}_2 \hat{\Psi}_{\text{ion}}^\dagger(\mathbf{R}_1) \hat{\Psi}_{\text{ion}}^\dagger(\mathbf{R}_2) \frac{Z^2 e^2}{|\mathbf{R}_1 - \mathbf{R}_2|} \hat{\Psi}_{\text{ion}}(\mathbf{R}_2) \hat{\Psi}_{\text{ion}}(\mathbf{R}_1)$$

while for the electrons we have:

$$T_{\text{el}} + V_{\text{el-el}} = \int d^3\mathbf{r} \hat{\Psi}_{\text{el}}^\dagger(\mathbf{r}) \left(-\frac{\hbar^2}{2m} \nabla_{\mathbf{r}}^2 \right) \hat{\Psi}_{\text{el}}(\mathbf{r}) + \frac{1}{2} \int \int d^3\mathbf{r}_1 d^3\mathbf{r}_2 \hat{\Psi}_{\text{el}}^\dagger(\mathbf{r}_1) \hat{\Psi}_{\text{el}}^\dagger(\mathbf{r}_2) \frac{e^2}{|\mathbf{r}_1 - \mathbf{r}_2|} \hat{\Psi}_{\text{el}}(\mathbf{r}_2) \hat{\Psi}_{\text{el}}(\mathbf{r}_1),$$

In the jellium model, we assume that the ions are fixed in space and that they form a uniform background of positive charge, while the electrons are free to move in this background, such that the density can be expressed as:

$$\rho_{jel}(\mathbf{R}) = \frac{N}{\Omega},$$

where N is the number of nuclei and Ω is the volume of the system, such is constant. In the end the idea is to get rid of the ionic part (the nuclei, which have constant density) and to focus only on the electronic part, which is the one that determines the properties of the system. Thus the ion-ion interaction can be expressed as

$$V_{\text{ion-ion}} = \frac{1}{2} \int \int d^3\mathbf{R}_1 d^3\mathbf{R}_2 \rho_{jel}(\mathbf{R}_1) \rho_{jel}(\mathbf{R}_2) \frac{1}{|\mathbf{R}_1 - \mathbf{R}_2|},$$

where we are basically multiplying the two infinitesimal charges $\rho_{jel}(\mathbf{R}_1) d^3\mathbf{R}_1$ and $\rho_{jel}(\mathbf{R}_2) d^3\mathbf{R}_2$ and then integrating over all space after dividing by the distance between the two charges. Thus we can write

$$\begin{aligned} V_{\text{ion-ion}} &= \frac{1}{2} \int \int d^3\mathbf{R}_1 d^3\mathbf{R}_2 \frac{N^2}{\Omega^2} \frac{1}{|\mathbf{R}_1 - \mathbf{R}_2|} \\ &= \frac{N^2}{2\Omega^2} \int d^3\mathbf{R} \int d^3\xi \frac{1}{|\xi|} = \frac{N^2}{2\Omega^2} \Omega \lim_{q \rightarrow 0} V_q, \end{aligned}$$

⁴Capital letters for nuclei coordinates and small letters for electrons coordinates.

where in the end we have performed the change of variables $\boldsymbol{\xi} = \mathbf{R}_1 - \mathbf{R}_2$ and we have expressed the integral in terms of the Fourier transform of the Coulomb potential, which diverges as $q \rightarrow 0$.

Moving on to the electron-ion interaction, we can express it as:

$$\begin{aligned} V_{\text{ion-el}} &= - \sum_{\sigma} \int \int d^3\mathbf{r} d^3\mathbf{R} \rho_{\text{el}}(\mathbf{r}) \rho_{\text{jel}}(\mathbf{R}) \frac{1}{|\mathbf{r} - \mathbf{R}|} \\ &= - \frac{N}{\Omega} \sum_{\sigma} \int \int d^3\mathbf{r} d^3\mathbf{R} \rho_{\text{el}}(\mathbf{r}) \frac{1}{|\mathbf{r} - \mathbf{R}|} = - \frac{N}{\Omega} \left[\sum_{\sigma} \int d^3\mathbf{r} \rho_{\text{el}}(\mathbf{r}) \right] \int d^3\boldsymbol{\xi} \frac{1}{|\boldsymbol{\xi}|} = - \frac{N^2}{\Omega^2} \lim_{q \rightarrow 0} V_q, \end{aligned}$$

so that in the end if we sum the ion-ion and ion-electron interactions we get:

$$V_{\text{ion-ion}} + V_{\text{ion-el}} = \frac{N^2}{2\Omega^2} \lim_{q \rightarrow 0} V_q - \frac{N^2}{\Omega^2} \lim_{q \rightarrow 0} V_q = - \frac{N^2}{2\Omega^2} \lim_{q \rightarrow 0} V_q,$$

which... check computations.

Moving onto the interesting term, the electron-electron interaction, we can express it as:

$$V_{\text{el-el}}(q=0) = \frac{1}{2} \sum_{\sigma, \sigma'} \sum_{\mathbf{k}_1, \mathbf{k}_2} \left[\lim_{q \rightarrow 0} V_q \right] \hat{a}_{\mathbf{k}_1, \sigma}^{\dagger} \hat{a}_{\mathbf{k}_2, \sigma'}^{\dagger} \hat{a}_{\mathbf{k}_2, \sigma'} \hat{a}_{\mathbf{k}_1, \sigma},$$

where now we are interested to change the order of the ladder operators, since we are dealing with fermions, so that the last term becomes:

$$\begin{aligned} \hat{a}_{\mathbf{k}_1, \sigma}^{\dagger} \hat{a}_{\mathbf{k}_2, \sigma'}^{\dagger} \hat{a}_{\mathbf{k}_2, \sigma'} \hat{a}_{\mathbf{k}_1, \sigma} &= - \hat{a}_{\mathbf{k}_1, \sigma}^{\dagger} \hat{a}_{\mathbf{k}_2, \sigma'}^{\dagger} \hat{a}_{\mathbf{k}_1, \sigma} \hat{a}_{\mathbf{k}_2, \sigma'} + \hat{a}_{\mathbf{k}_1, \sigma}^{\dagger} \hat{a}_{\mathbf{k}_1, \sigma} \hat{a}_{\mathbf{k}_2, \sigma'} \hat{a}_{\mathbf{k}_2, \sigma'}^{\dagger} \\ &= - \hat{a}_{\mathbf{k}_1, \sigma}^{\dagger} \hat{a}_{\mathbf{k}_1, \sigma} \hat{a}_{\mathbf{k}_2, \sigma'}^{\dagger} \hat{a}_{\mathbf{k}_2, \sigma'} + 2 \hat{a}_{\mathbf{k}_1, \sigma}^{\dagger} \hat{a}_{\mathbf{k}_1, \sigma} \hat{a}_{\mathbf{k}_2, \sigma'} \hat{a}_{\mathbf{k}_2, \sigma'}^{\dagger} \\ &= - n_{\mathbf{k}_1, \sigma} n_{\mathbf{k}_2, \sigma'} + 2 n_{\mathbf{k}_1, \sigma}, \end{aligned}$$

so that the electron-electron interaction can be expressed as:

$$V_{\text{el-el}}(q=0) = \frac{1}{2\Omega} \sum_{\sigma, \sigma'} \sum_{\mathbf{k}_1, \mathbf{k}_2} \left[\lim_{q \rightarrow 0} V_q \right] (\hat{n}_{\mathbf{k}_1, \sigma} \hat{n}_{\mathbf{k}_2, \sigma'} - \delta_{\mathbf{k}_1, \mathbf{k}_2} \delta_{\sigma, \sigma'} \hat{n}_{\mathbf{k}_1, \sigma}),$$

and remembering that the sum of the occupation numbers is equal to the total number of particles $\sum_{\mathbf{k}} \hat{n}_{\mathbf{k}} = N$, we can simplify the expression further

$$V = \frac{1}{2\Omega} (N^2 - N) \lim_{q \rightarrow 0} V_q \sim \frac{N^2}{2\Omega} \lim_{q \rightarrow 0} V_q,$$

showing how the electron-electron interaction diverges as $q \rightarrow 0$, which is a consequence of the long-range nature of the Coulomb interaction. This divergence is a problem for the theory, but if we sum this contribution with the ion-ion and ion-electron interactions, we get the following hamiltonian for the jellium model:

$$H_{\text{el}} = \sum_{\mathbf{k}\sigma} \frac{\hbar^2 k^2}{2m} \hat{a}_{\mathbf{k}, \sigma}^{\dagger} \hat{a}_{\mathbf{k}, \sigma} + \frac{1}{2\Omega} \sum_{\sigma, \sigma'} \sum_{\mathbf{k}_1, \mathbf{k}_2, \mathbf{q} \neq 0} \left(\frac{4\pi e^2}{q^2} \right) \hat{a}_{\mathbf{k}_1 + \mathbf{q}, \sigma}^{\dagger} \hat{a}_{\mathbf{k}_2 - \mathbf{q}, \sigma'}^{\dagger} \hat{a}_{\mathbf{k}_2, \sigma'} \hat{a}_{\mathbf{k}_1, \sigma},$$

which is the Hamiltonian of the electron gas, where we have neglected the constant term that diverges as $q \rightarrow 0$, since it does not affect the properties of the system. This Hamiltonian is the starting point for many-body perturbation theory and other techniques used to study interacting systems, and it is a very important model in condensed matter physics, since it captures many of the essential features of real materials, such as metals and semiconductors.

The strength of this model is that it is simple enough to be solved exactly in some cases, but it is also rich enough to capture many of the essential features of real materials, such as the Fermi surface, the density of states, and the response functions. It is also a very useful model to study the effects of interactions on the properties of the system, such as the formation of collective excitations, the screening of the Coulomb interaction, and the emergence of new phases of matter.

TODO: change all the ladder operators from a to c, since we are using electrons.

TODO: correct computations

Appendices

A | Notation and Conventions

B | Computations and further developments